

We've Got Your Nose; Solving Problems Using Computer Aided Design At Weir ESCO

Melinda Tran

WEIR

ESCO



Weir ESCO

- From 1871, Weir ESCO manufactures ground engagement tools for construction, mining, and industrial applications
- MEOT team
 - Works on the points, adapters, and shrouds



N5-TS640H Adapter

Hensley Nose



Problem

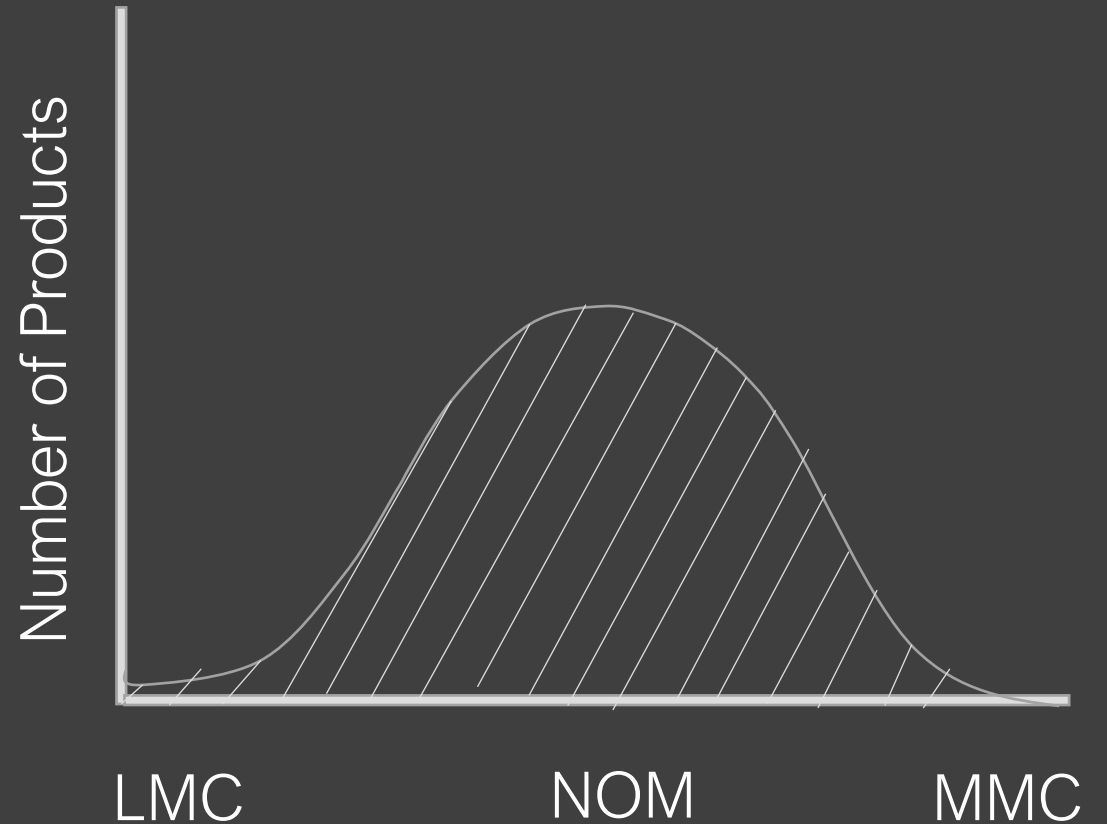
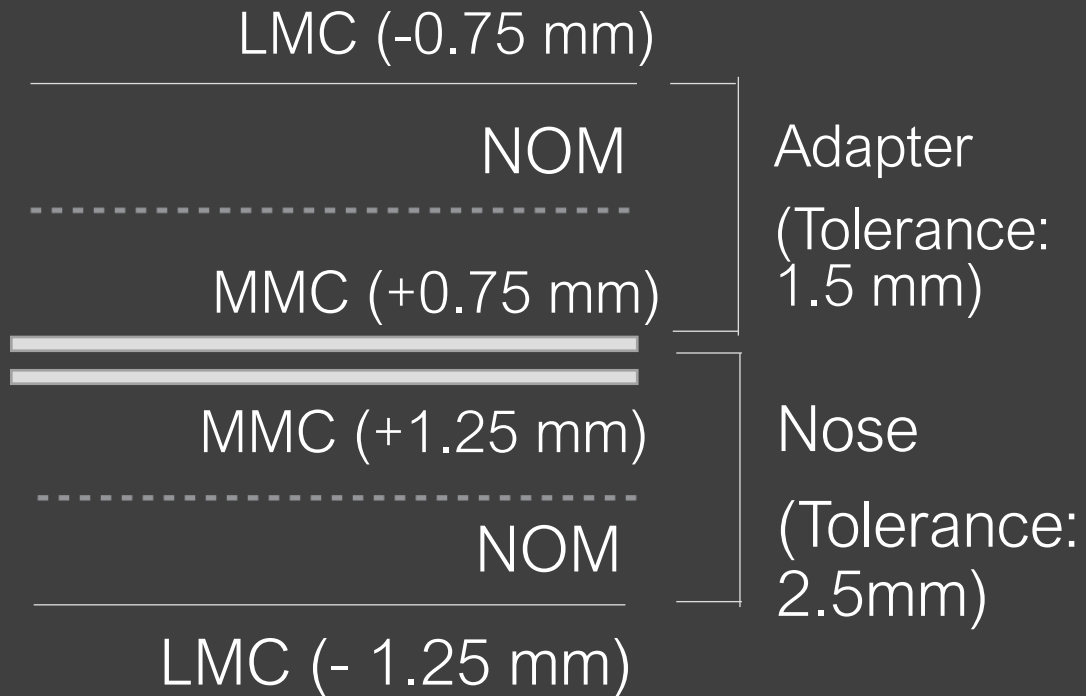
The Hensley nose (competitor's nose) fit to the N5-TS640H adapter was loose

Materials

LMC = Least Material Condition

NOM = Nominal

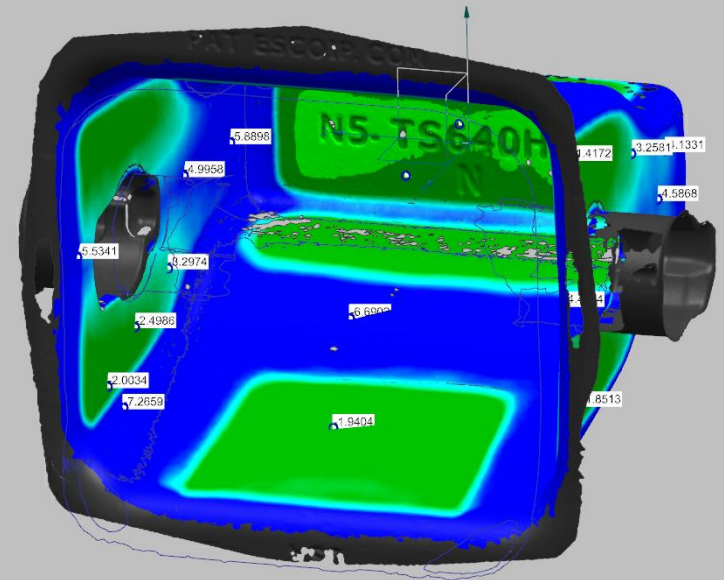
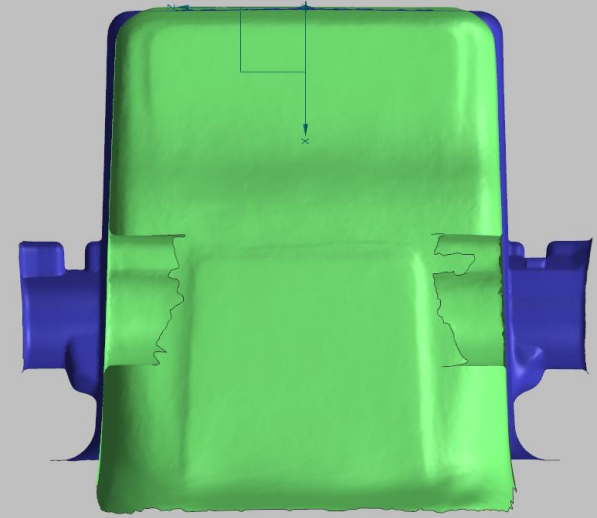
MMC = Maximum Material Condition

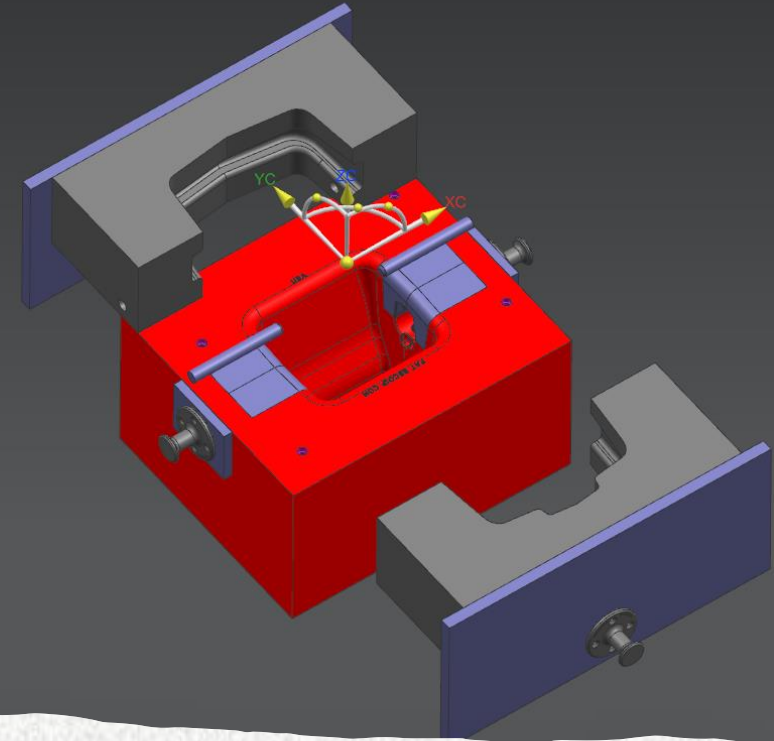
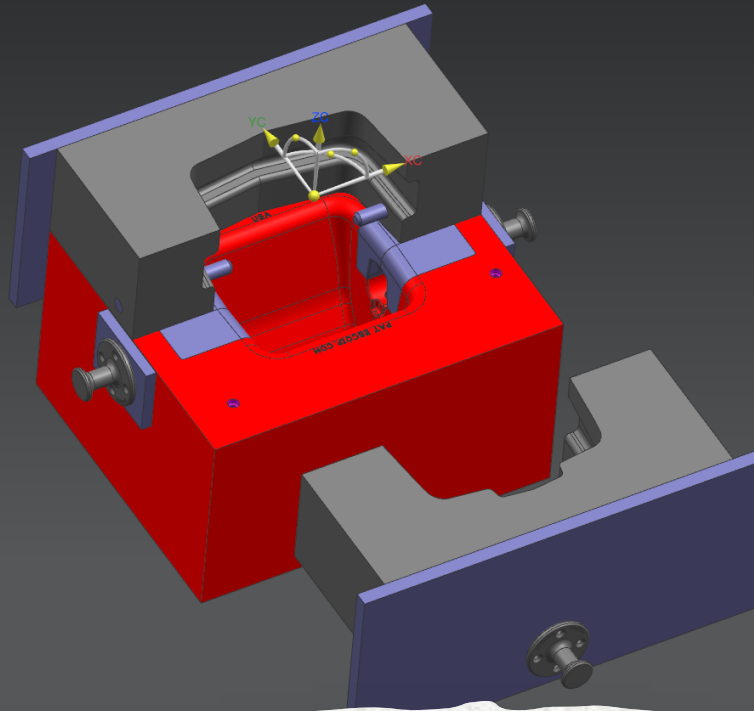
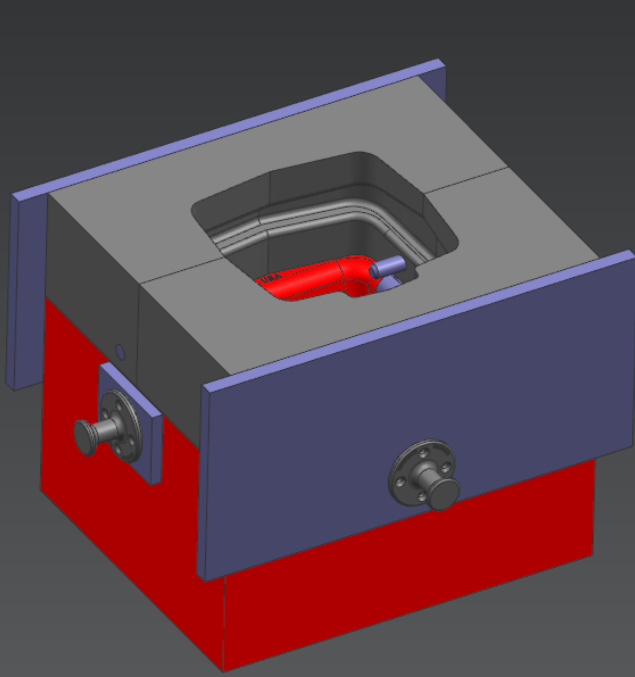


Measurements

Measuring the gap on CAD

- 1) Average Nose scans and Cavity scans on PolyWorks
- 2) Align Average scans on PolyWorks
- 3) Employ deviations tool to see spacing on NX



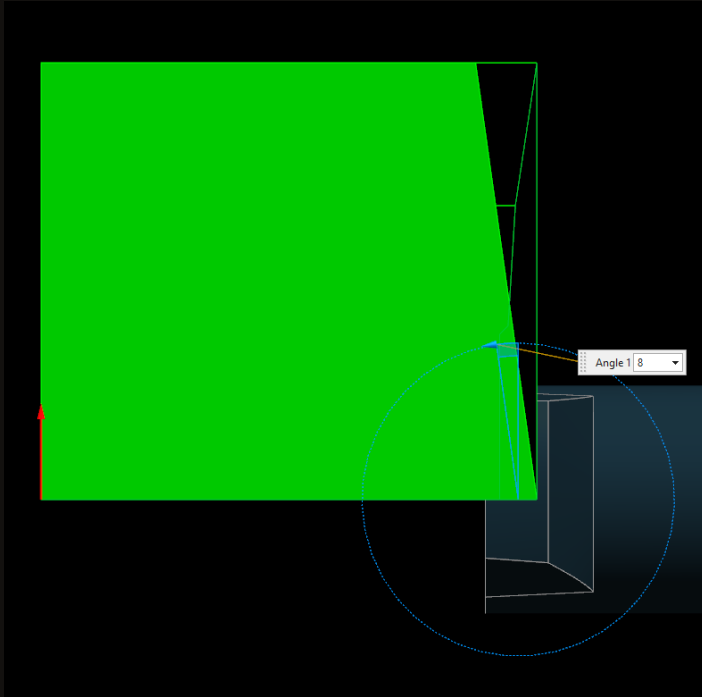


Tooling Process

Must take into account the process capability of the manufacturing process

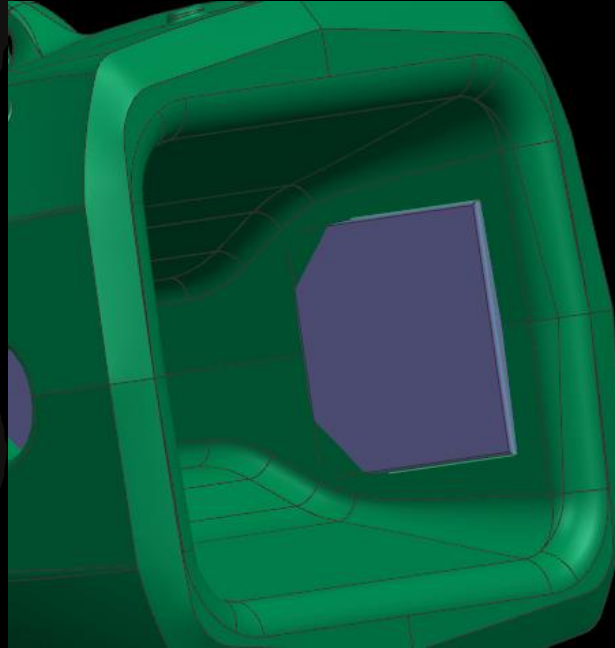
Planning

Goal: Decrease the spacing to 1.5 mm (Ideal spacing)



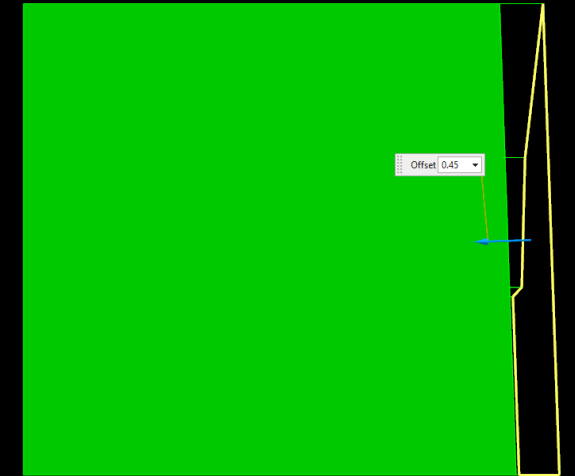
Editing draft

- Changing angle of side



Adding fit pads

- Adding extra pieces of metal in casting process

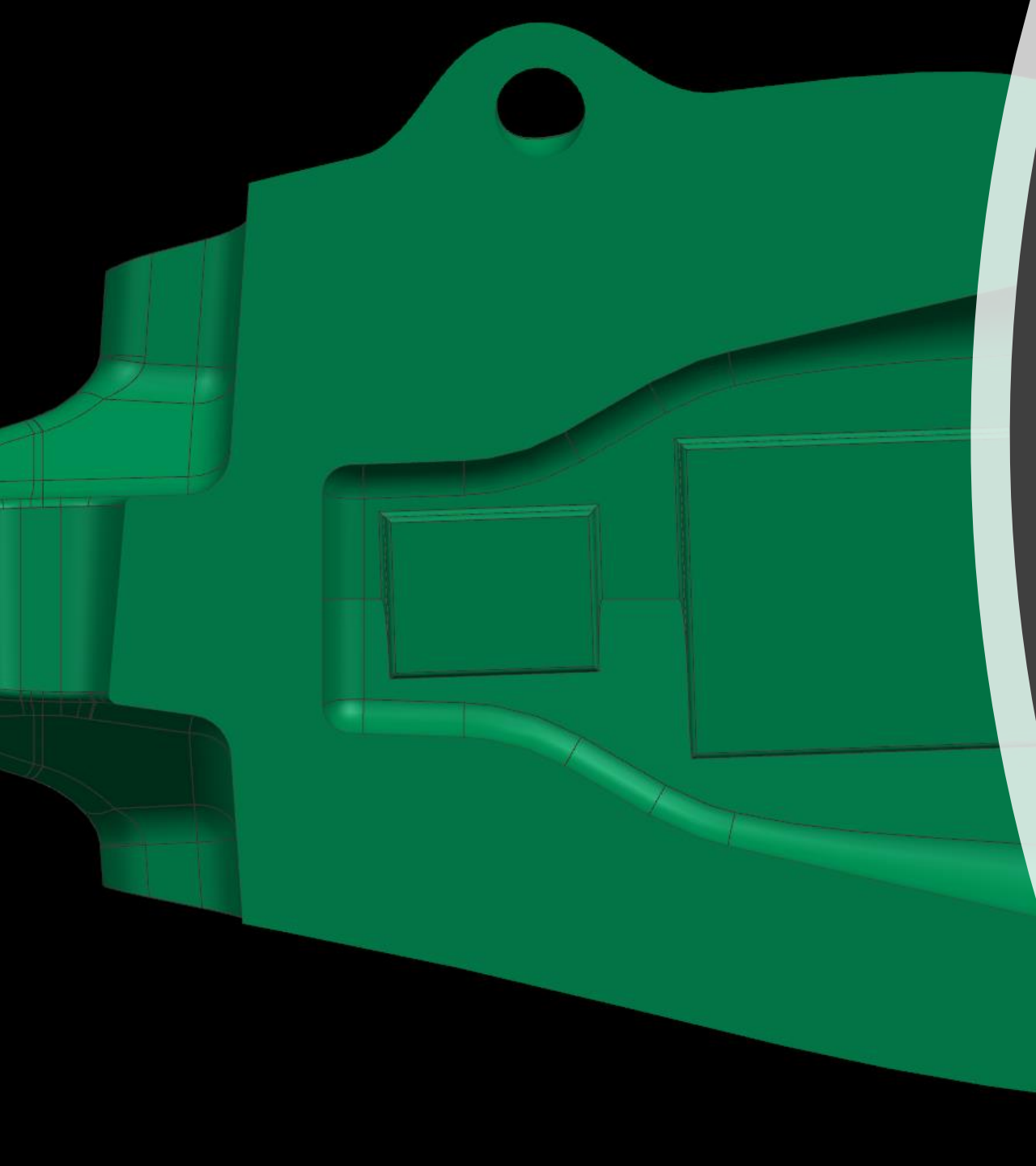


Offsetting the side

- Moving side in and out



Solutions



Fit Pads

- Keeping draft at 2 degrees
- Adding fit pads in the front at 4.5 mm
- Adding fit pads in the back at 0.5 mm

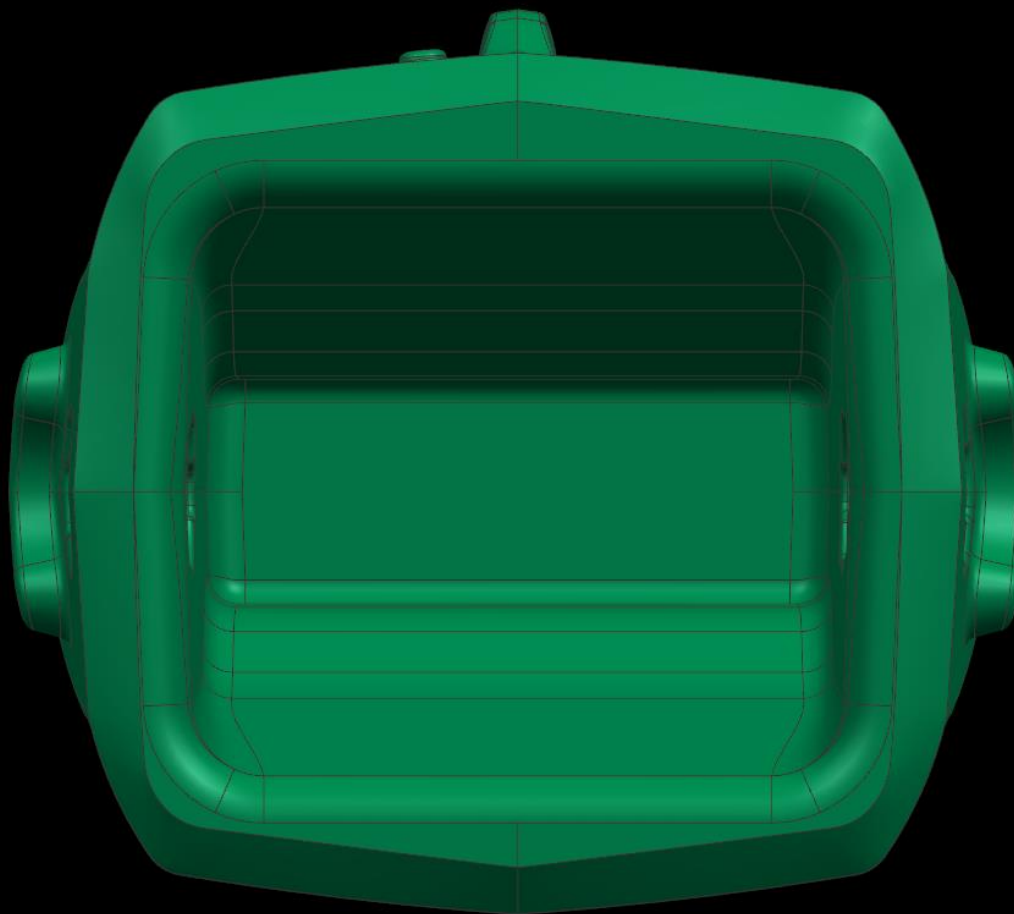
Fit Pads + Draft

- Subtracting draft to 0 degrees
- Adding fit pads in the front at 4.5 mm
- Adding fit pads in the back at 0.5 mm



Draft + Offset Face

- Removing draft to 0 degrees
- Offsetting face closer by 0.5



A large excavator bucket is the central focus, resting on a dirt surface. In the background, two construction workers wearing hard hats and safety vests are standing near the bucket. The scene is dimly lit, suggesting an overcast day or a shaded area. The excavator's arm and other parts of the machine are visible in the background.

Future

For the company:

- Implement idea into production
- Continue to expand products inventory

For me:

- Utilize knowledge to inform future career plans

Acknowledgments

Thank you for this experience!

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Saturday Academy & Weir ESCO



Questions ?